

Harisankar G

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SUMMARY

AI and Machine Learning graduate skilled in Deep Learning, Computer Vision, Predictive Analytics, and Data Science. Experienced in developing machine learning models, real-time AI applications, and data-driven solutions using Python, PyTorch, TensorFlow, SQL.

EDUCATION

Vellore Institute of Technology
Integrated MTech in Artificial Intelligence

Oct 2022 – Present
GPA: 8.22/10

TECHNICAL SKILLS

Programming: Python, R, SQL

Machine Learning: CNNs, LSTM, Predictive Modeling, Time-Series Forecasting, Hyperparameter Tuning, RAG

Deep Learning: Py Torch, TensorFlow, U-Net, Computer Vision, Medical Image Segmentation

Libraries: scikit-learn, Pandas, NumPy, OpenCV, YOLOv8, MediaPipe

Software Development: REST APIs, Backend Systems, Data Pipelines, Model Evaluation

Cloud & Tools: AWS (EC2, S3), Docker, Git

PROJECTS

ML-Based Threat Detection in Crowded Environments — Python, Computer Vision [GitHub](#)

- Designed a real-time video analytics pipeline using YOLOv8 object detection, MediaPipe pose estimation, and OpenCV for anomaly detection across multiple threat categories.
- Engineered spatial and temporal feature extraction modules from human pose keypoints and movement trajectories to support multi-class activity recognition and behavior analysis.
- Built a scalable modular inference pipeline for real-time processing and achieved **99.1% AUC** and **92% F1-score**; work was published as an IEEE research paper.

AI-Driven Smart Storage Optimization Engine — Python, scikit-learn, SQLite [GitHub](#)

- Developed an intelligent storage optimization engine utilizing BLAKE3 hashing, entropy analysis, and Content Defined Chunking (CDC) techniques to identify duplicate and inefficient file storage patterns.
- Built machine learning models using engineered file-system features to recommend optimal compression strategies and improve storage utilization through data-driven decision making.
- Implemented SQLite-backed metadata management, similarity search, drift detection, and an interactive analytics dashboard for monitoring storage efficiency and optimization outcomes.

Brain Tumor Segmentation and Visualization System — Python, PyTorch, U-Net [GitHub](#)

- Developed an end-to-end deep learning pipeline for automatic brain tumor segmentation from MRI scans using U-Net architecture trained on more than 57K MRI image slices from the BraTS dataset.
- Implemented preprocessing, augmentation, training, evaluation, and inference workflows using PyTorch with combined Dice + BCE loss functions, achieving a Dice score of **0.9284**.
- Built a modular codebase with custom dataset loaders, evaluation metrics, prediction pipelines, and visualization utilities for reproducible experimentation and model interpretation.

INTERNSHIP

Data Analyst Intern — Zidio

Aug 2025 – Oct 2025

- Built LSTM-based predictive models for financial time-series datasets, developing preprocessing pipelines using Pandas and NumPy to clean, transform, and integrate raw data from multiple sources.
- Automated data preparation and model evaluation workflows using Python scripts, maintaining data quality validation with RMSE and MAE metrics.

Web Development Intern — TISE

Oct 2025 – Dec 2025

- Built backend modules for structured data storage and retrieval using relational databases.
- Applied validation logic and data consistency checks; contributed to debugging and feature updates in production code.

ACHIEVEMENTS

IEEE Publication: *Machine Learning-Based Threat Detection in Crowded Environments*

Amazon ML Challenge 2025: Ranked **192nd** globally

UIDAI Data Hackathon 2026: Finalist

CERTIFICATIONS

Applied Machine Learning in Python — University of Michigan, Coursera

2023

Google Cloud Generative AI Virtual Internship — Google Cloud

2025